MONASH University

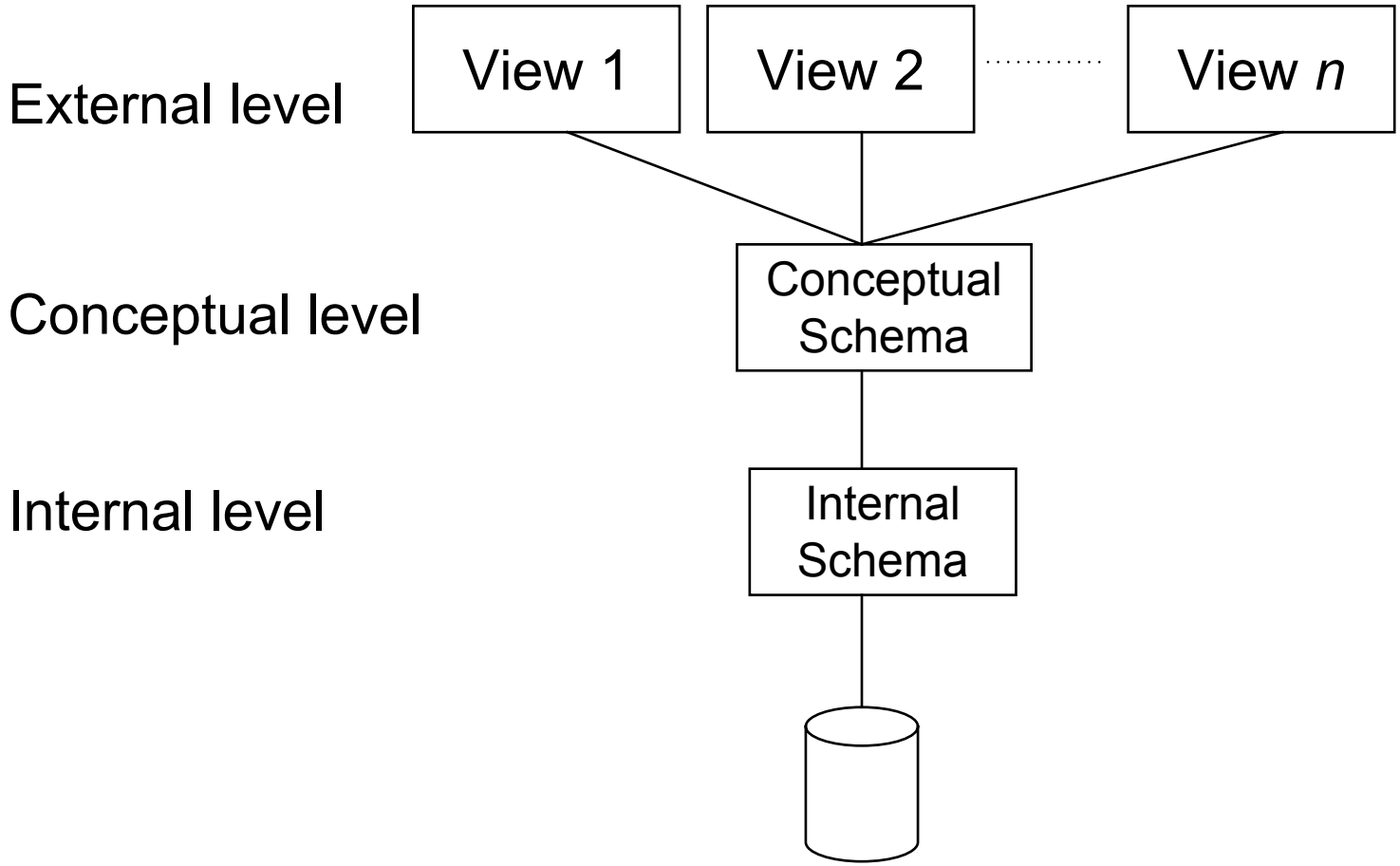
MONASH
INFORMATION
TECHNOLOGY

Database Design 1: Conceptual Modelling

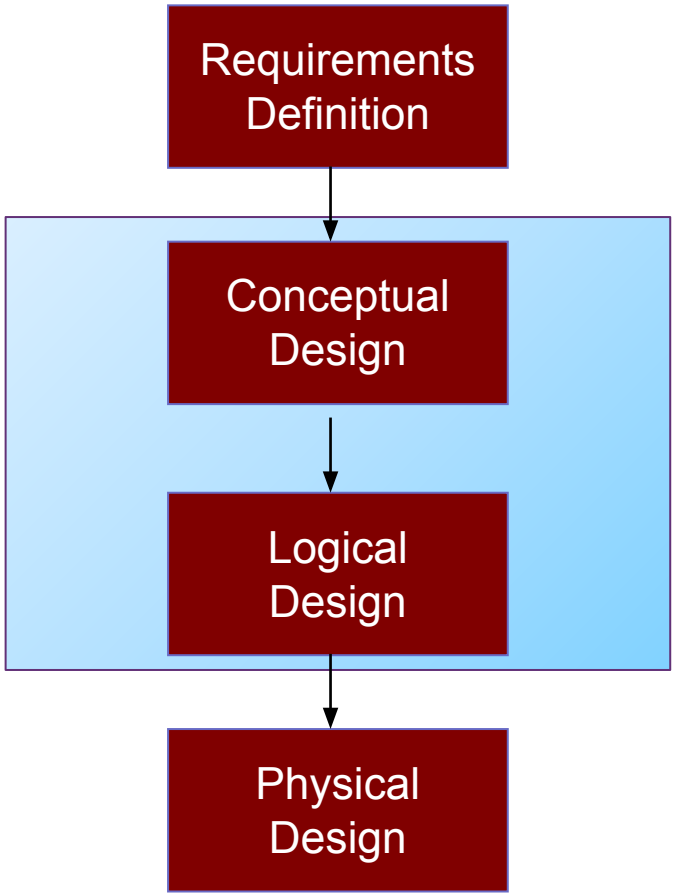
Lindsay Smith

GROUP OF EIGHT AUSTRALIA

ANSI/SPARC architecture



The Database Design Life Cycle



Requirements Definition

- Identify and analyse user views.
- A 'user view' may be a report to be produced or a particular type of transaction that should be supported.
- Corresponds to the external level of the ANSI/SPARC architecture.
- Output is a statement of specifications which describes the user views' particular requirements and constraints.

Student view

Web Enrolment System

- Enrolment / Re-Enrolment
- Fees / Scholarships
- Student Services
- Course Progression
- Enrolment Access Dates
- WES Guides
- WES Survey

Monash Links

- my.monash
- Allocate+ (Class Allocation)
- MUTTS (Class Timetable)
- Moodle

Web Enrolment System

✔ Check Faculty and Course requirements before enrolling. It is your responsibility to ensure your enrolment complies with course requirements. Arrange a meeting with your Course Adviser if you are unsure.

✔ You are required to enrol in all units that you expect to study in each Semester. Need help? Click here

Remember to Submit after making any changes.
Then wait for your Transaction Number.

Unit Enrolment form

Any *Unconfirmed* units are core requirements of your course. Click on *Enrol* in the Action column to confirm enrolment in these units.

Course Location:	PARKVILLE(**FORCED** - What does this mean?)					
	To Add Units Click here					
Unit code	Action	Unit name	Campus	Semester	Type	Credits
PAC1111	Enrol	Introduction to physiology **UNCONFIRMED**	PAR	Semester 1 (2014)	DAY	--
PAC1121	Enrol	Bioorganic and medicinal chemistry I **UNCONFIRMED**	PAR	Semester 1 (2014)	DAY	--
PAC1211	Enrol	Physicochemical basis of pharmacy **UNCONFIRMED**	PAR	Semester 1 (2014)	DAY	--
PAC1311	Enrol	Pharmacy, health and society I **UNCONFIRMED**	PAR	Semester 1 (2014)	DAY	--
PAC1132	Enrol	Systems physiology **UNCONFIRMED**	PAR	Semester 2 (2014)	DAY	--
PAC1142	Enrol	Bioorganic and medicinal chemistry II **UNCONFIRMED**	PAR	Semester 2 (2014)	DAY	--
PAC1222	Enrol	Drug delivery I **UNCONFIRMED**	PAR	Semester 2 (2014)	DAY	--
PAC1322	Enrol	Pharmacy, health and society II **UNCONFIRMED**	PAR	Semester 2 (2014)	DAY	--
					Total credits:	0

If you do not get a Transaction Number after you submit, your enrolment is not complete.

Staff and Student View

Unit guides

MONASH University

Home | Find a unit guide |

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PDF

FIT9132: Introduction to databases

Semester 1 2016

Contents

- Unit handbook information
  - Synopsis
  - Mode of delivery
  - Workload requirements
  - Unit relationships
    - Prerequisites
    - Prohibitions
    - Co-requisites
  - Chief Examiner
    - Campus Lecturer(s)

Unit handbook information

Synopsis

This unit will introduce the concept of data management in an organisation through relational database technology. Theoretical foundation of relational model, analysis and design, implementation of relational database using SQL will be covered.

Mode of delivery

Caulfield (Evening)
Caulfield (Online)

Workload requirements

Minimum total expected workload equals 12 hours per week comprising:

(a.) Contact hours for on-campus students:

Admin View

Allocate+

Subject Administrator
FIT9132_CA_S2_DAY, INTRO TO DATABASES

Activity Groups:

Show Subject | Special Consideration | Update Details | Section Insert

Lecture (Lecture)
(READ ONLY)

Enrolments:	302
Preferences:	128
Allocations:	302
Seats Provided:	328

PASS_Session (PASS_Session)
(READ ONLY)

Enrolments:	302
Preferences:	0
Allocations:	51
Seats Provided:	48

Warning: Not enough seats provided

Studio (Studio)
(READ ONLY)

Enrolments:	302
Preferences:	144
Allocations:	302
Seats Provided:	329

FIT9132_CA_S2_DAY:INTRO TO DATABASES (Studio)

Allocate | Add Activity | Show Message | Allocated List | Unallocated List | Bul

Functions	Activity Code	Campus	Day	Start Time	Location	Staff	Duratio
Delete | Edit | List | Constraint | Context | Email	01	CA	Wed	14:00	CA_B/B345	Monika Schwarz	120
Delete | Edit | List | Constraint | Context | Email	02	CA	Wed	14:00	CA_B/B343	Mark Creado	120
Delete | Edit | List | Constraint | Context | Email	03	CA	Wed	16:00	CA_B/B342B	Minh Vuong	120
Delete | Edit | List | Constraint | Context | Email	04	CA	Wed	14:00	CA_B/B342B	Minh Vuong	120
Delete | Edit | List | Constraint | Context | Email	05	CA	Wed	14:00	CA_B/B346B	Paras Sitoula	120
Delete | Edit | List | Constraint | Context | Email	06	CA	Wed	16:00	CA_B/B343	Angela Zhang	120
Delete | Edit | List | Constraint | Context | Email	07	CA	Wed	17:30	CA_B/B344	Mark Creado	120
Delete | Edit | List | Constraint | Context | Email	08	CA	Thu	10:00	CA_B/B346	Lindsay Smith	120
Delete | Edit | List | Constraint | Context | Email	09	CA	Thu	14:00	CA_B/B350	Karan Pedramrazi	120
Delete | Edit | List | Constraint | Context | Email	10	CA	Thu	16:00	CA_B/B345	Minh Vuong	120
Delete | Edit | List | Constraint | Context | Email	11	CA	Wed	17:30	CA_K/K107	Paras Sitoula	120
Delete | Edit | List | Constraint | Context | Email	12	CA	Fri	14:00	CA_B/B346	Manoj Kathpalia	120

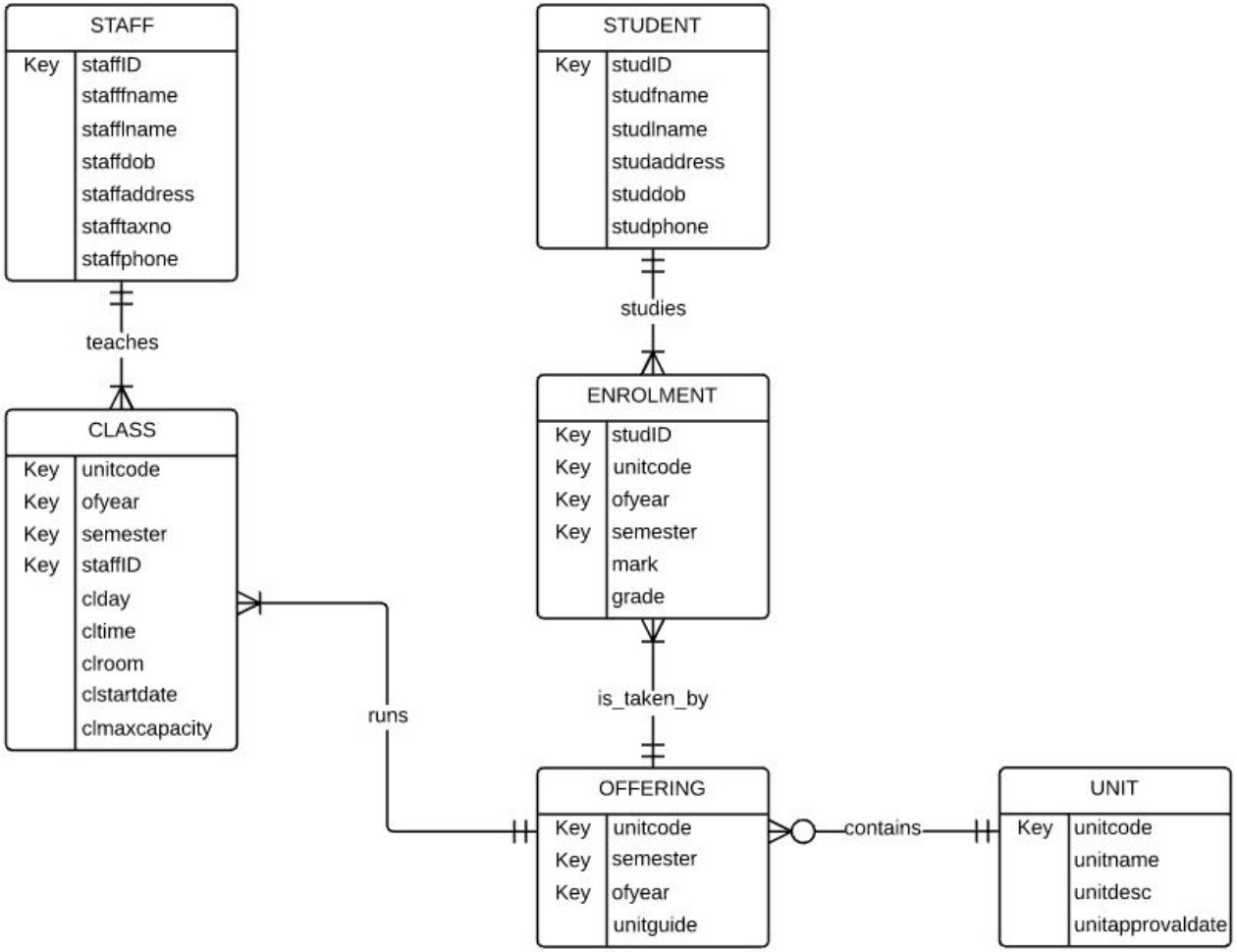
ER Modeling

- ER (Entity-Relationship) model developed by Peter Chen in 1976 to aid database design.
- May be used for conceptual (ERD)/logical design (ERD like).
- ER diagrams give a visual indication of the design.
- Basic components:
  - Entity
  - Attribute
  - Relationship

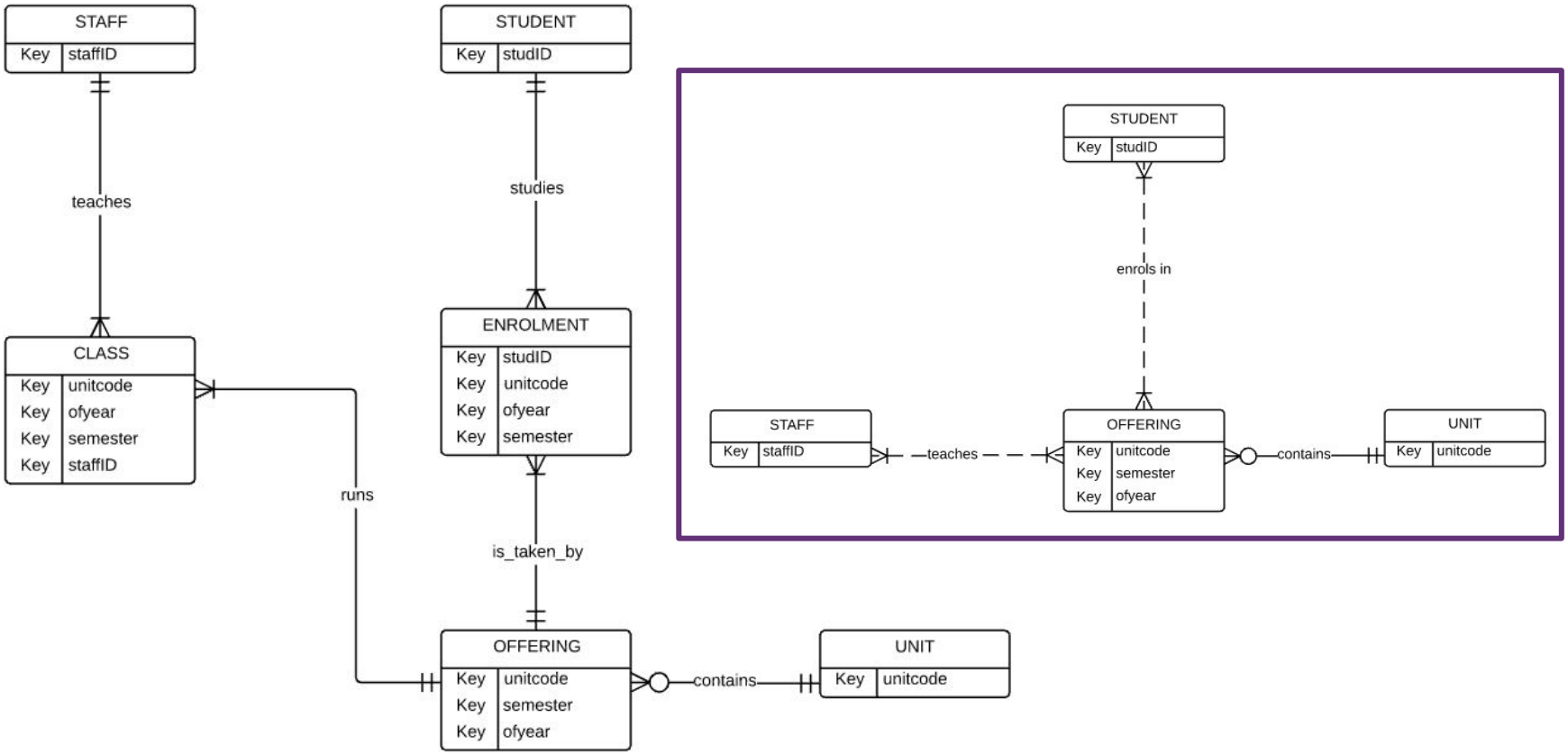
Conceptual Design

- Develop the enterprise data model.
- Corresponds to the conceptual level of the ANSI/SPARC architecture.
- Independent of all physical implementation considerations.
- Various design methodologies may be employed, including the ER (Entity-Relationship) approach.

Conceptual Level (ER Model)



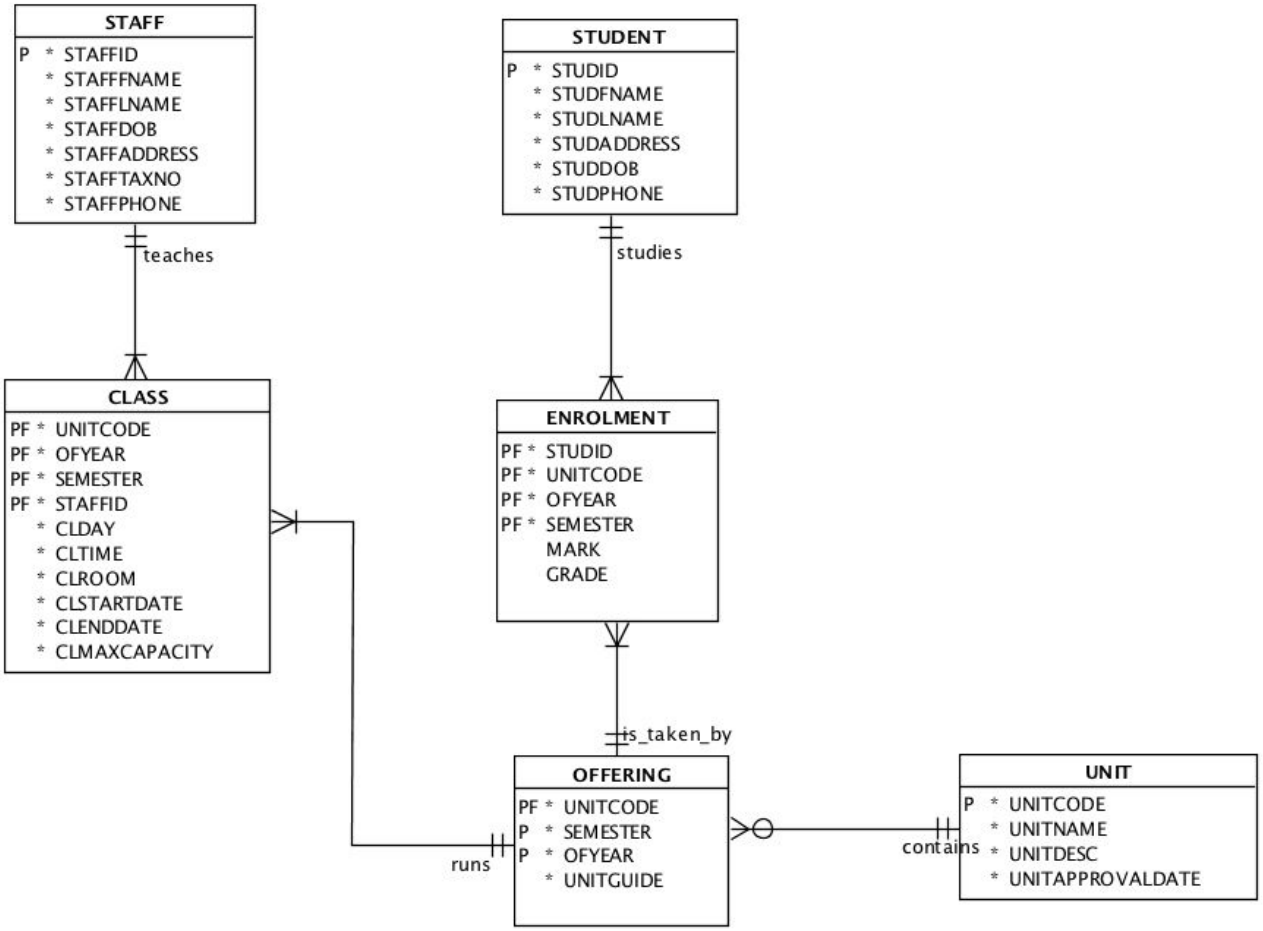
Conceptual Level (ER Model) - Keys only



Logical Design

- Develop a data model which targets a particular database model (e.g. relational, hierarchical, network, object-oriented).
- Independent of any implementation details which are specific to any particular DBMS package.
- Normalisation technique (see week 5) is used to test the correctness of the logical model.
- May also be considered to correspond to the conceptual level of the ANSI/SPARC architecture.

Logical Level (Logical Model)



Physical Design

- Develop a strategy for the physical implementation of the logical data model.
- Choose appropriate storage structures, indexes, file organisations and access methods which will most efficiently support the user requirements.
- Physical design phase is dependent on the particular DBMS environment in use.
- ANSI/SPARC internal level.
- Shown in SQL Developer Data Modeller as the Relational Model

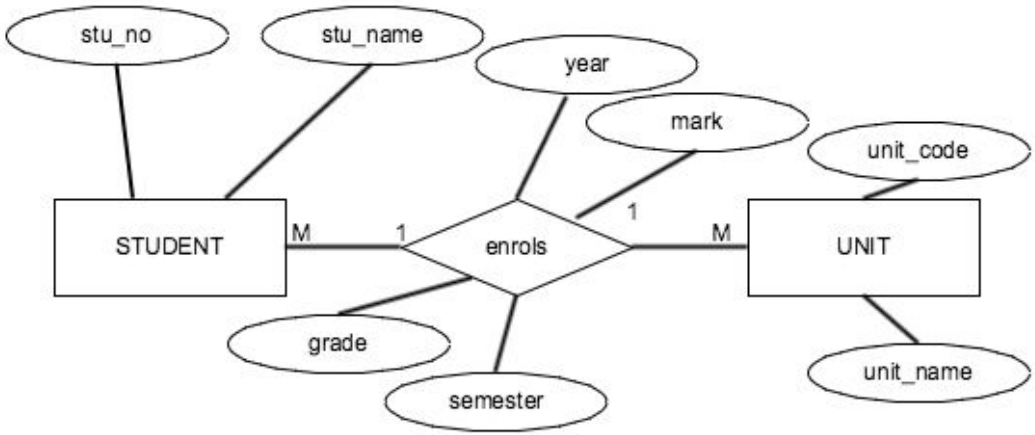
Physical Level – Starting point

Oracle Database 11g | Relational_1 | Generate | Clear

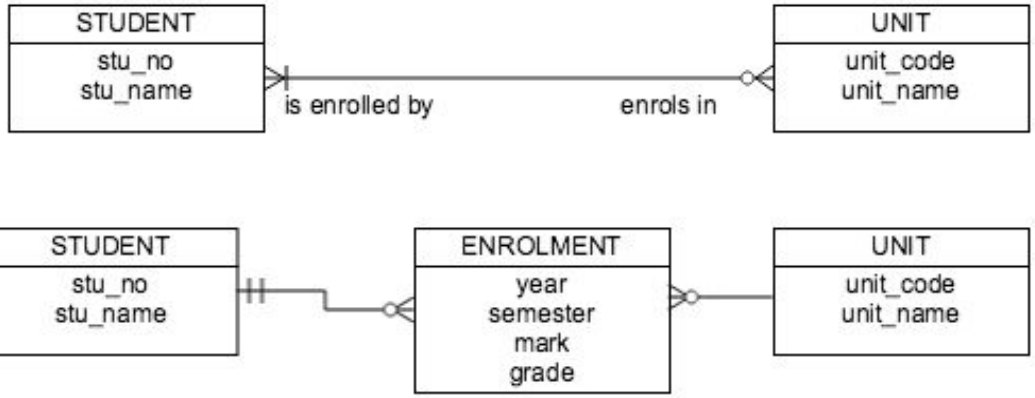
```
9  create
10   table enrolment
11   (
12     unitcode char (10) not null ,
13     semester number (1) not null ,
14     ofyear   date not null ,
15     studid   number (10) not null ,
16     mark     number (3) ,
17     grade    char (2)
18   ) ;
19
20
21 alter table enrolment add constraint enrol_mark_chk check (mark between 0 and
22 100) ;
23
24 alter table enrolment add constraint enrol_grade_chk check (grade in ('N','P',
25 'C','D','HD')) ;
26
27 alter table enrolment add constraint enrol_pk primary key ( semester, ofyear,
28 studid, unitcode ) ;
29
30 create
31   table offering
32   (
33     unitcode  char (10) not null ,
34     semester  number (1) not null ,
35     ofyear    date not null ,
36     chiefexam number (10) not null
37   ) ;
38
39 alter table offering add constraint semester_chk check (semester between 1 and
40 3) ;
41
42
43 alter table offering add constraint offering_pk primary key ( unitcode,
44 semester, ofyear ) ;
45
46 create
47   table prereq
48   (
49     unitcode      char (10) not null
```

ENTITY RELATIONSHIP DIAGRAM

ERD - Notation



Chen



Information Engineering/James Martin/Crows foot

ERD – Notation cont'd

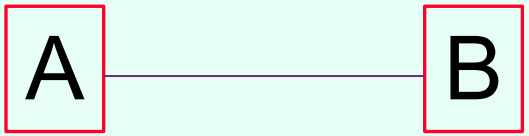
Chen's Notation

- Semantically rich.
- Complex diagram.
- 'Pure' conceptual level.

Information Engineering

- Less semantics.
- Simpler diagram.
- Mix between conceptual and logical levels.

Entity, Attributes and Relationships



Please note this diagram is incomplete

Q1. How many entities are there in the above diagram?

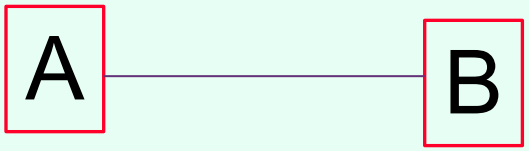
A. 1

B. 2

C. 3

D. 4

mars.mu
Feed Code: HYT80D



Please note this diagram is incomplete

Q2. How many relationship are there on the above diagram?

What is the degree of the relationship (the number of entities participating in the relationship) ?

A. 1, unary
B. 2, binary
C. 1, binary
D. 3, ternary

Q3. "An employee is assigned to be a member of a team. A team with more than 5 members will have a team leader. The members of the team elect the team leader."

List the entity(s) which you can identify in the above statement. Enter them one by one - select the tick for each and then use the X to add a second entity if required.

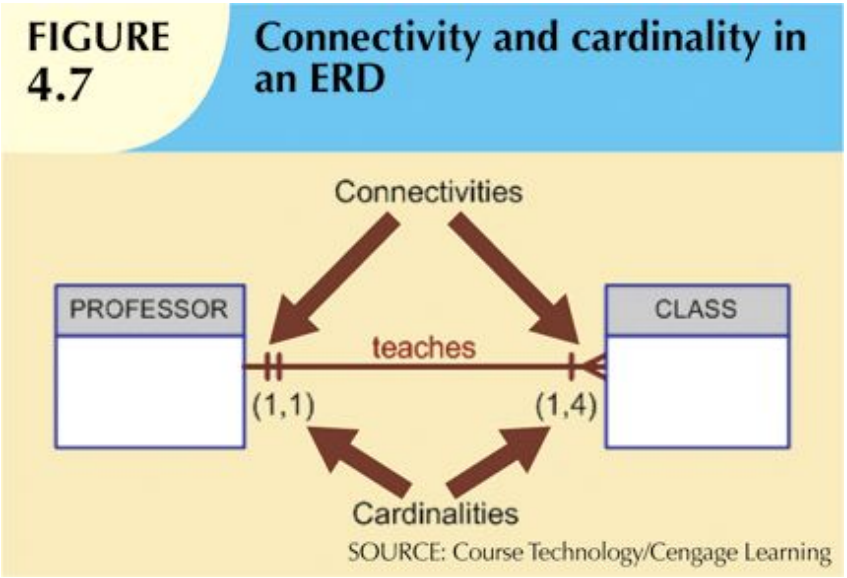
Q4. How many relationships connect TEAM and EMPLOYEE?

A. 1

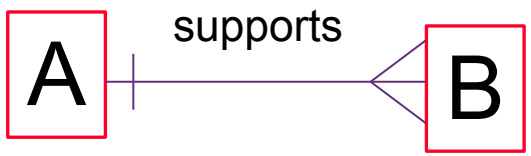
B. 2

C. 3

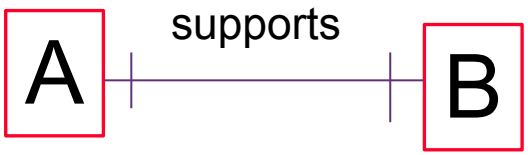
D. 4



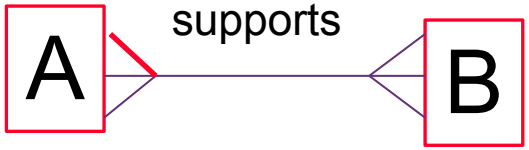
one to many



one to one



many to many



Relationship Participation

Q5. "A Team with more than 5 members will have a team leader. The members of the team elect the team leader."

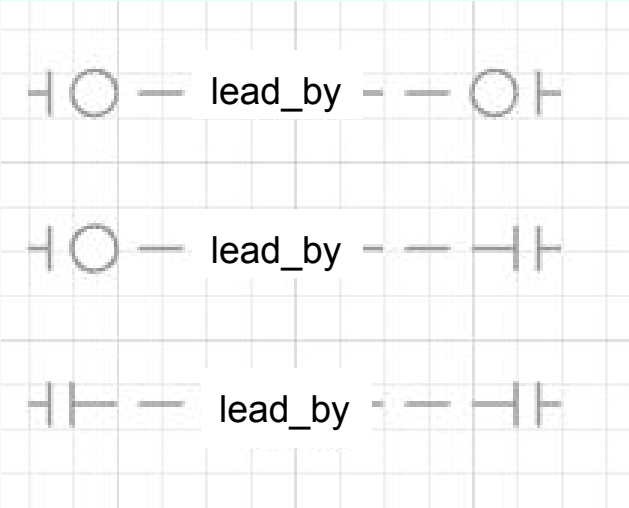
What is the relationship participation of the relationship

TEAM --- lead_by --- EMPLOYEE

A.

B.

C.

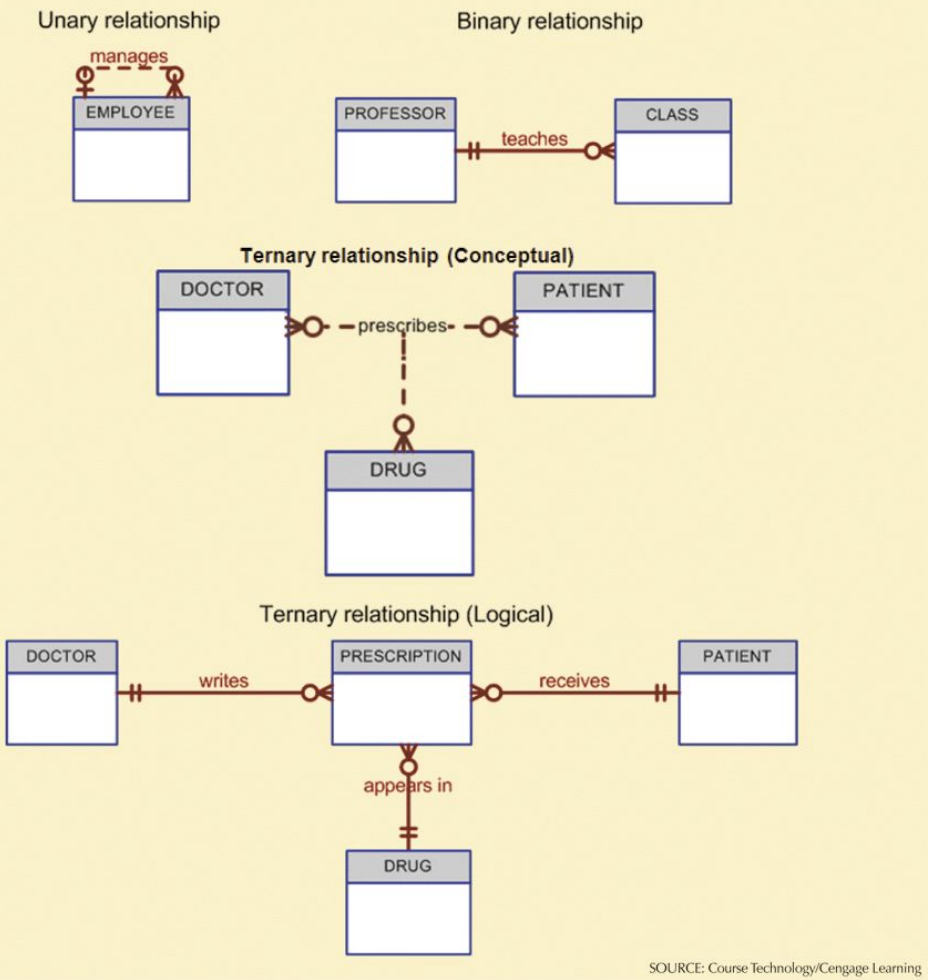


Q6. "The company also introduced a mentoring program, whereby a new employee will be paired with someone who has been in the company longer."

How many entity/ies do you need to model the mentoring program?

A. 1
B. 2
C. 3

FIGURE 4.15 Three types of relationship degree



SOURCE: Course Technology/Cengage Learning

Q7. "To attract high calibre talent, the company provides generous remuneration package as well as health insurance support for the employees and their family. To do this, the HR team needs to know the details of the family members. The family information will be recorded"

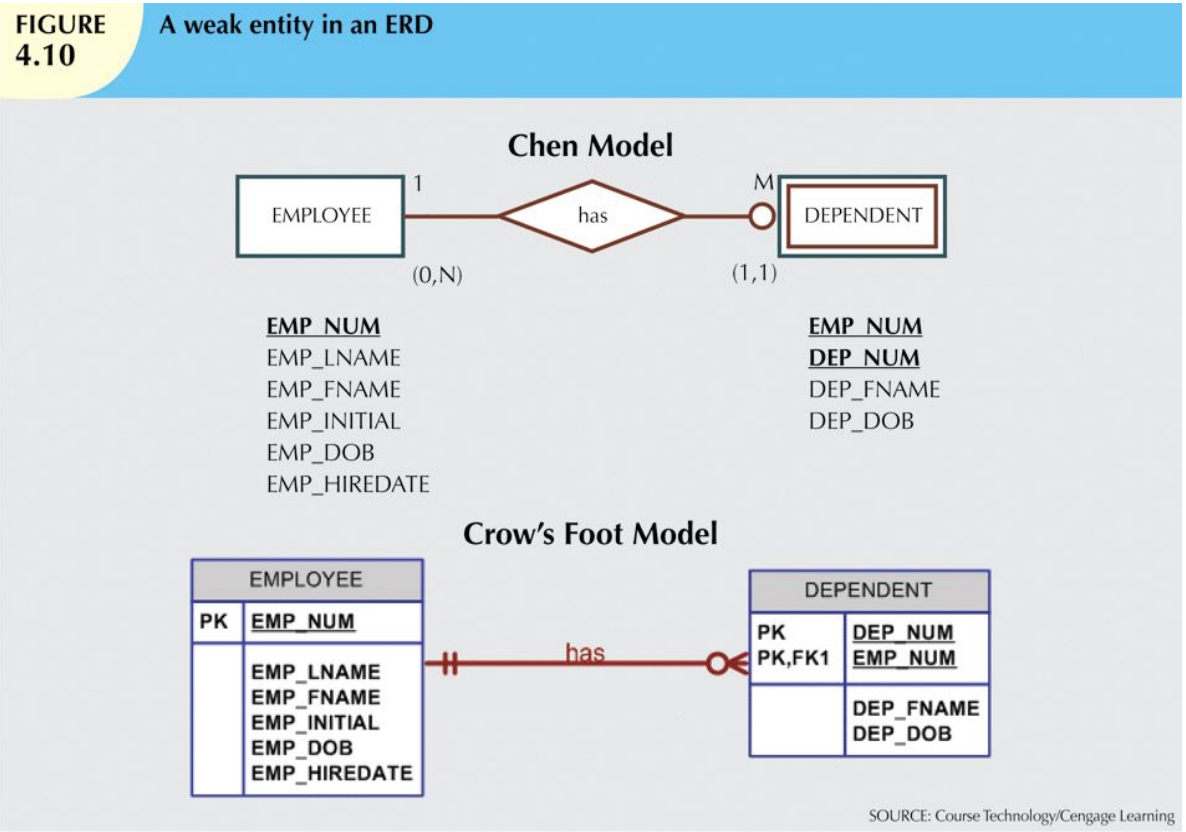
Choose a TRUE statement.

A. EMPLOYEE entity is a strong entity and FAMILY is a strong entity.
B. EMPLOYEE entity is a weak entity and FAMILY is a strong entity.
C. EMPLOYEE entity is a strong entity and FAMILY is a weak entity.
D. EMPLOYEE entity is a strong entity and FAMILY is strong entity.

Weak vs Strong Entity

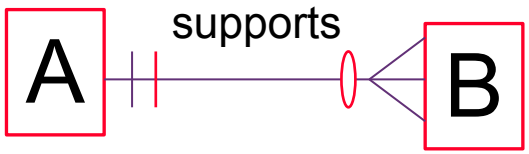
- Strong entity
  - Has a key which may be defined without reference to other entities.
  - For example EMPLOYEE entity.
- Weak entity
  - Has a key which requires the existence of one or more other entities.
  - For example FAMILY entity - need to include the key of employee to create a suitable key for family
- Database designer often determines whether an entity can be described as weak based on business rules
  - customer pays monthly account
    - Key: cust_no, date_paid, or
    - Key: payment_no (surrogate? – not at conceptual level)

Weak vs Strong Entity



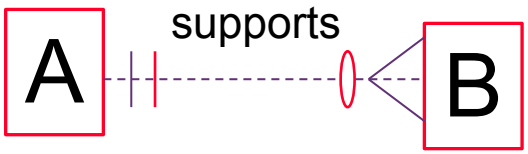
Identifying vs Non-Identifying Relationship

- **Identifying**
  - Identifier of A is part of identifier of B.



  - Shown with solid line
  - Enrolment's PK includes student id, which is an identifier of student.

- **Non-identifying**
  - Identifier of A is NOT part of identifier of B.



  - Shown with broken line
  - Department no (identifier of department) is not part of Employee's identifier.

Types of Attributes

- Simple
  - Cannot be subdivided
  - Age, sex, marital status
- Composite
  - Can be subdivided into additional attributes
  - Address into street, city, zip
- Single-valued
  - Can have only a single value
  - Person has one social security number
- Multi-valued
  - Can have many values
  - Person may have several college degrees
- Derived
  - Can be derived with algorithm
  - Age can be derived from date of birth

Q8. The employee details that will be recorded are: Employee number, Full name, Address, Date of birth, Tax file number and Skill(s). Examples of skills are Java, Python, UNIX, Relational db, Mongo db, etc

Choose a TRUE statement.

A. ADDRESS is a multi-valued attribute and SKILL is a multi-valued attribute.

B. ADDRESS is a composite attribute and SKILL is a composite attribute.

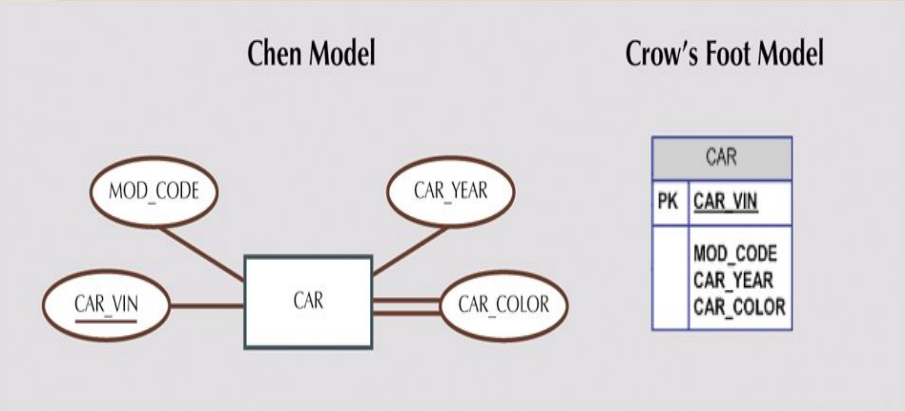
C. ADDRESS is a composite attribute and SKILL is a multi-valued attribute.

D. ADDRESS is a multi-valued attribute and SKILL is a composite attribute.

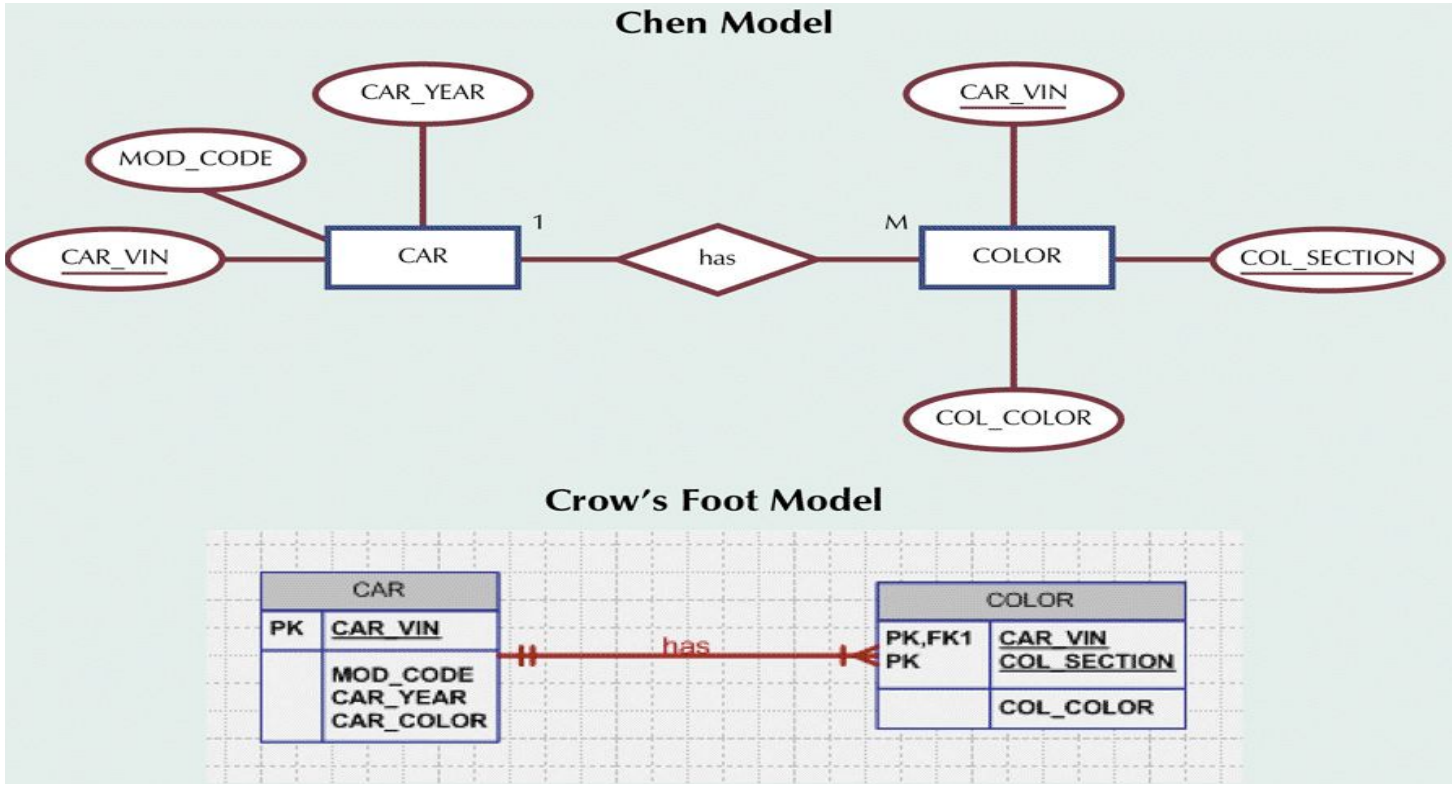
Multivalued Attribute

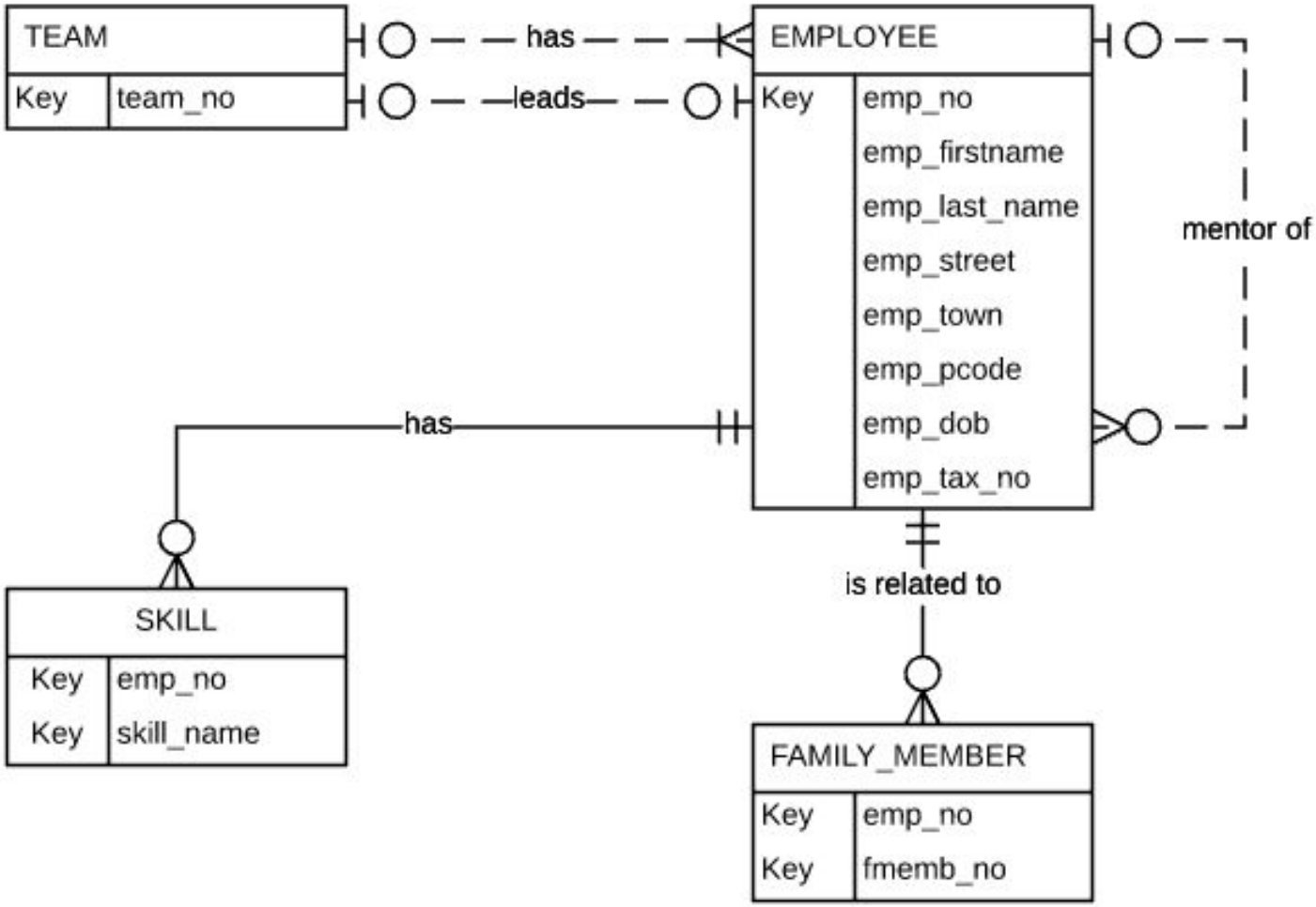
- An attribute that has a list of values.
- For example:
  - Car colour may consist of body colour, trim colour, bumper colour.
- Crow's foot notation does not support multivalued attributes. Values are listed as a separate attribute.

FIGURE 4.3 A multivalued attribute in an entity



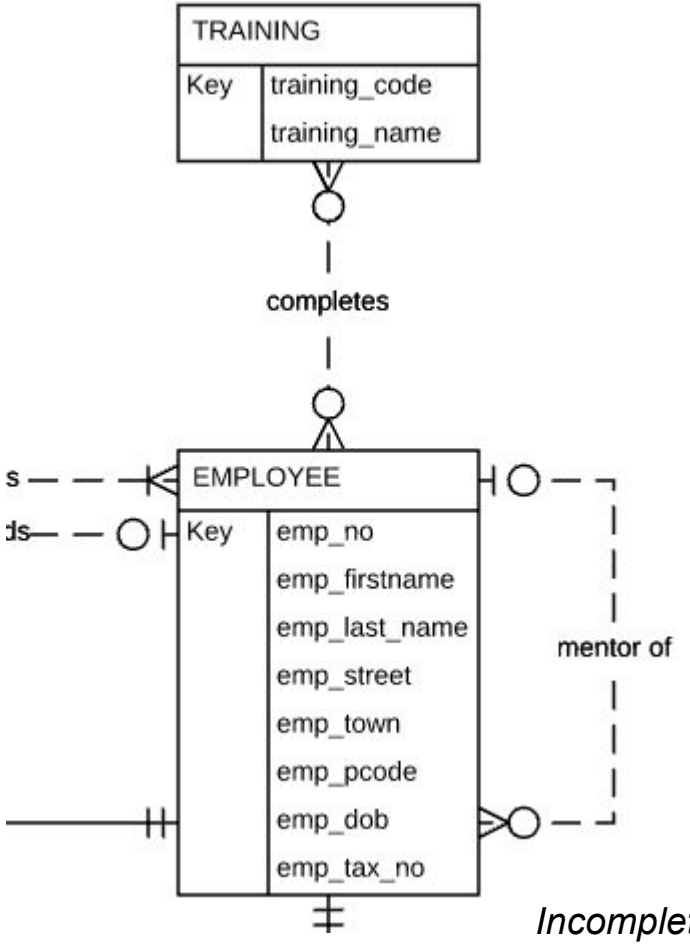
Resolving Multivalued Attributes




TEAM
Key team_no
has
leads
EMPLOYEE
Key emp_no
emp_firstname
emp_last_name
emp_street
emp_town
emp_pcode
emp_dob
emp_tax_no
mentor of
has
SKILL
Key emp_no
Key skill_name
is related to
FAMILY_MEMBER
Key emp_no
Key fmemb_no

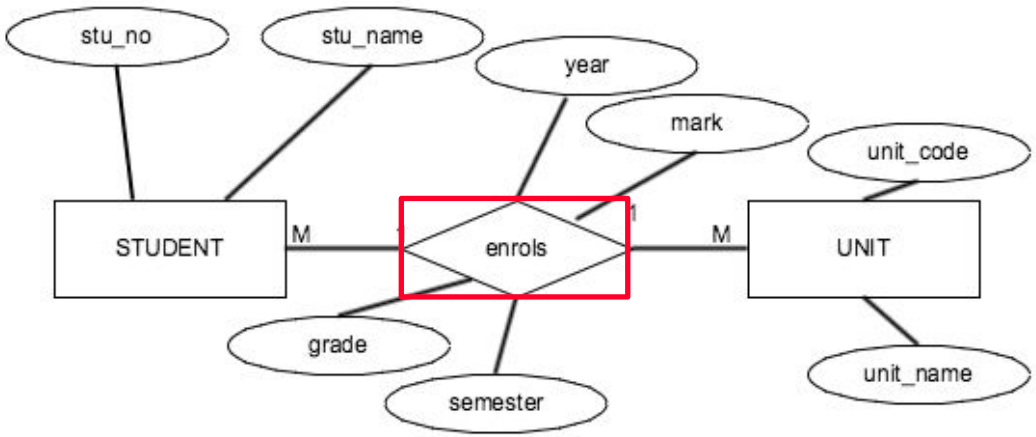
..."the company provides several in-house training programs. The HR team needs to keep track of the details about who has done what. An employee can do several training programs.

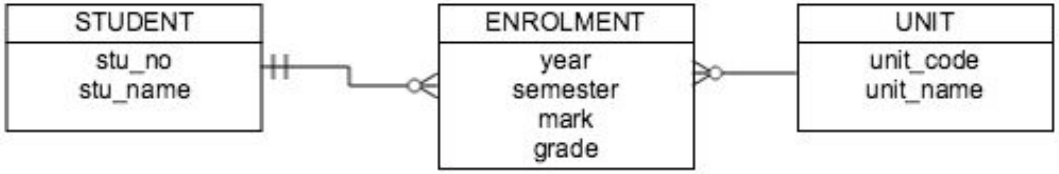
At the completion of a training, a certificate will be provided to the employee containing the training name and the completion date."...

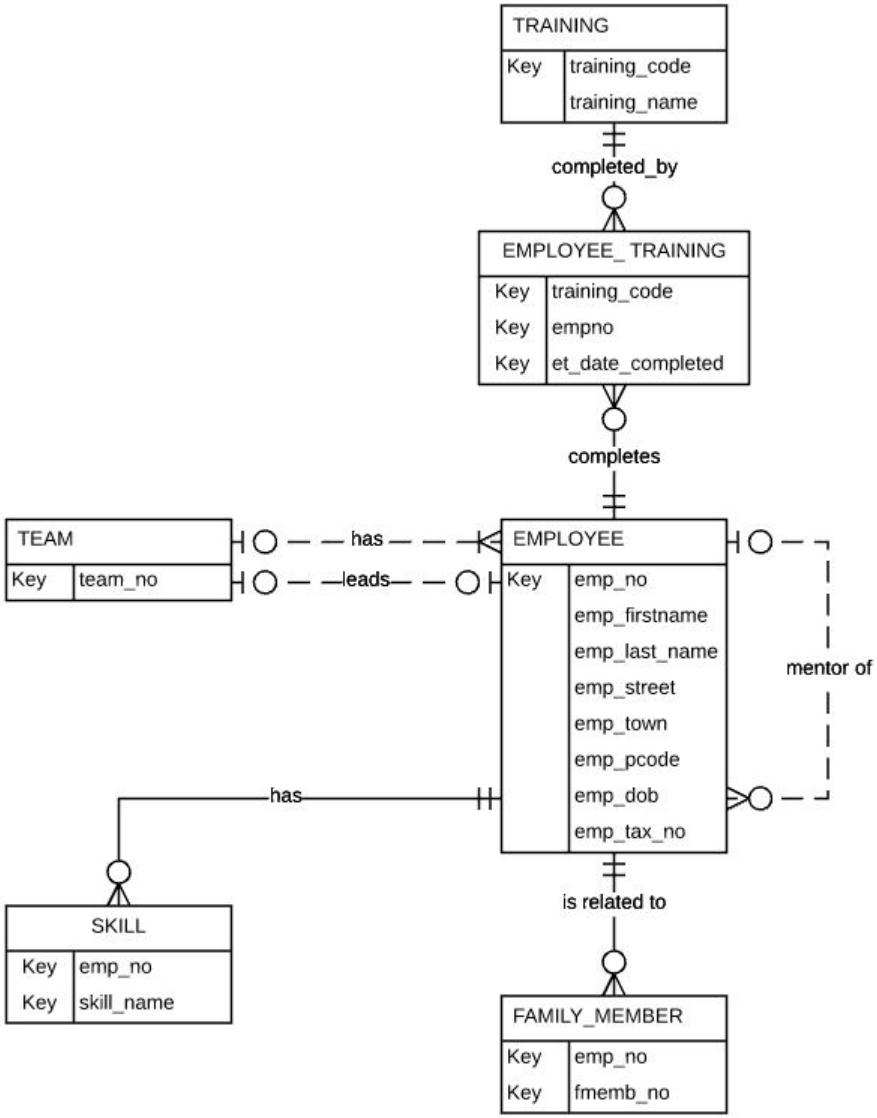


Incomplete model

Associative (or Composite) Entity






TRAINING
Key training_code
training_name
completed_by
EMPLOYEE_ TRAINING
Key training_code
Key empno
Key et_date_completed
completes
TEAM
Key team_no
has
leads
EMPLOYEE
Key emp_no
emp_firstname
emp_last_name
emp_street
emp_town
emp_pcode
emp_dob
emp_tax_no
mentor of
has
SKILL
Key emp_no
Key skill_name
is related to
FAMILY_MEMBER
Key emp_no
Key fmemb_no